

EXPERIMENT-04

AIM

To write a Python program using NumPy to calculate the total yearly sales and determine the percentage increase in sales from the first quarter to the fourth quarter using quarterly sales data stored in a NumPy array.

ALGORITHM

1. Start the program.
2. Import the NumPy library.
3. Create a NumPy array `sales_data` containing four elements representing quarterly sales.
4. Calculate the total yearly sales using `np.sum()`.
5. Extract the first quarter and fourth quarter sales.
6. Compute the percentage increase using the formula:

$$\text{Percentage Increase} = \frac{Q4 - Q1}{Q1} \times 100$$

7. Display the total yearly sales and percentage increase.
8. End the program.

PROGRAM

```
import numpy as np

# Quarterly sales data: Q1, Q2, Q3, Q4
sales_data = np.array([45000, 52000, 58000, 70000])

total_sales = np.sum(sales_data)

q1 = sales_data[0]
q4 = sales_data[3]
```

```
percentage_increase = ((q4 - q1) / q1) * 100  
  
print("Total yearly sales:", total_sales)  
  
print("Percentage increase from Q1 to Q4:", percentage_increase)
```

SAMPLE INPUT

```
[45000, 52000, 58000, 70000]
```

SAMPLE OUTPUT

```
Total yearly sales: 225000  
Percentage increase from Q1 to Q4: 55.55555555555556
```

RESULT

The program successfully calculated the total yearly sales and computed the percentage increase in sales from the first quarter to the fourth quarter using NumPy operations.